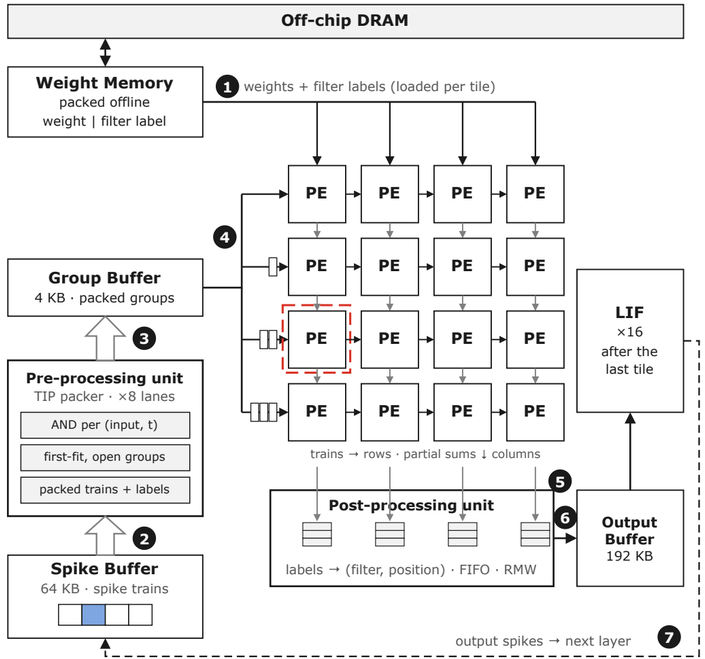
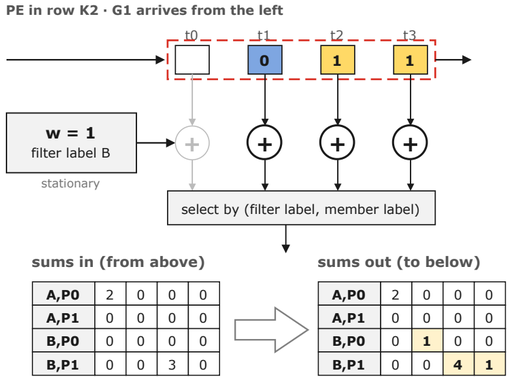


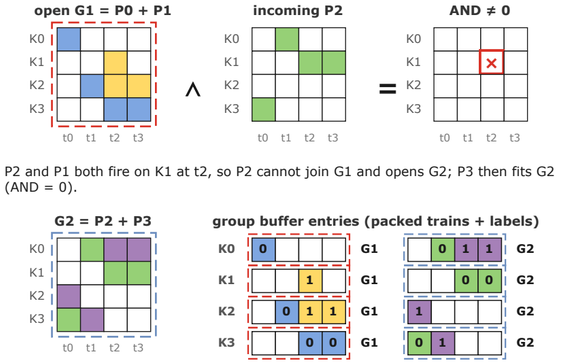
(a) TIDES: offline-packed weights, run-time TIP in the pre-processing unit, 4×4 PEs with T adders each



Adder t adds w only when bit t is 1. The label picks the member: +1 to (B, P0) at t1, +1 to (B, P1) at t2 and t3. All other partial sums pass through unchanged; the PE never compares or searches.

Per PE
1 weight + 2-bit filter label (shadow copy for the next tile)
T adders, one per timestep
F_{max}·G_{max}·T partial sums carried down the column

(b) Label-indexed PE: one weight serves all T timesteps of a packed train in one cycle

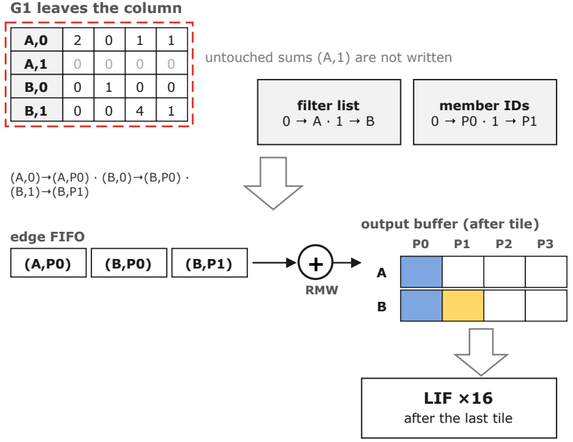


P2 and P1 both fire on K1 at t2, so P2 cannot join G1 and opens G2; P3 then fits G2 (AND = 0).

+ member IDs per group: G1 = {P0, P1}, G2 = {P2, P3} (7 bits each)

Hardware
8 lanes, each scanning one tile; per position: a R·T-bit mask, Q = 4 AND checks against the open groups, OR-merge on a fit
A group leaves when it holds G_{max} members or when the oldest open group is evicted
Silent positions are dropped before the array

(c) Pre-processing unit: one AND per (input, timestep) mask, first-fit into open groups



Hardware
4 FIFOs × 80 entries, one per column; 4 read-modify-writes per column per cycle into the 192 KB output buffer
Labels (2 bits) + per-group member IDs and per-column filter lists give the address (n, m); no search

(d) Post-processing unit: labels restore (filter, position); the output buffer sums the tiles

spike of P0 · spike of P1 · spike of P2 · spike of P3 · 0 · group G1 / G2 · 1 updated sum